HW7 for CS 6210

You may (and probably should) talk about problems with the each other, with the TA, and with me, providing attribution for any good ideas you might get. Your final write-up should be your own.

```
using LinearAlgebra
```

```
using Plots
```

1. Hessenberg Walker

Consider a Markov chain on $1, \ldots, n$ where

$$P(X_{t+1} = i | X_t = j) \propto \begin{cases} \lambda^{j-i}, & i \leq j+1 \\ 0, & \text{otherwise} \end{cases}$$

The upper Hessenberg transition matrix for the Markov chain is computed below (the routine `Pwalker`). Write an $O(n^2)$ time algorithm to compute the stationary distribution for any ergodic Hessenberg Markov chain (this implies the stationary distribution is nonzero on all nodes), and demonstrate it for this chain. Your code should have the signature

```
p = stationary_H(P)
```

The equation $\pi^\top P = \pi^\top$ rearranges to $\pi^\top (P - I) = 0^\top$, which is a homogeneous underdetermined system of rank $n-1$ (since $\mathbf{rank}(P - I) = n - 1$ for an ergodic chain). This has a one-dimensional nullspace, and we resolve the degree of freedom using the constraint $\sum_{i=1}^{n} \pi_i = 1$.

To form a full-rank system, we replace one row of $A = P - I$ (which is also upper Hessenberg) with the all-ones row $\mathbf{1}^\top$, and set the corresponding right-hand side entry in b to 1 (others 0). This yields the augmented system $A\pi = b$,

For non-replaced rows: $(P - I)$ equations hold (≈ 0).

For the replaced row r: $\sum \pi_i = 1$.

The choice of row r (1 or n) is based on comparing $|\mathbf{diag}(A)_1|$ and $|\mathbf{diag}(A)_n|$, replacing the row with the smaller absolute diagonal value. This is a stability heuristic: Gaussian elimination (GE) without pivoting uses diagonal entries as pivots, and replacing the smallest $|A_{k,k}|$ avoids a potentially tiny pivot, reducing backward error in floating-point arithmetic (especially for ill-conditioned P when $\lambda \gg 1$ or $\lambda \ll 1$, where π concentrates at boundaries). We then apply GE to triangularize A into upper triangular form, followed by back-substitution. Due to the Hessenberg structure: In column k,

only $A_{k+1,k}$ (subdiagonal) needs elimination (if not the replaced row). The replaced row (if below k) is also eliminated separately. Updates are confined to columns $j \geq k+1$ (no fill-in beyond the structure).

For an upper Hessenberg matrix, GE simplifies each column k has at most one nonzero below the diagonal ($A_{k+1,k}$). Eliminating it requires subtracting a multiple of row k from row $k+1$, updating only the $O(n-k)$ entries to the right ($j = k+1$ to n). No fill-in occurs (the matrix stays Hessenberg). Over $k = 1$ to $n-1$, total work is $\sum_{k=1}^{n-1} O(n-k) = O(n^2)$. Back-substitution is $O(n^2)$ in the worst case (summing $O(n-i)$ terms per row i), but for this structured upper triangular system, it's $O(n^2)$ total.

```
Pwalker (generic function with 1 method)
```

```
function Pwalker(λ, n)
    P = zeros(n,n)
    for j = 1:n
        imax = min(j+1,n)
        P[1:imax,j] .= λ.^-(0:imax-1) * (1-1/λ) / (1-λ^-imax)
    end
    P
end
```

```
stationary_H (generic function with 1 method)
```

```
function stationary_H(P)
    n = size(P, 1)
    A = copy(P)
    for i in 1:n
        A[i, i] -= 1
    end
    diag1 = abs(A[1, 1])
    diagn = abs(A[n, n])
    replace_row = diag1 < diagn ? 1 : n
    b = zeros(n)
    b[replace_row] = 1.0
    A[replace_row, :] .= 1.0
    for k in 1:n-1
        if k + 1 != replace_row
            m = A[k + 1, k] / A[k, k]
            A[k + 1, k] = 0.0
            for j in k + 1:n
                A[k + 1, j] -= m * A[k, j]
            end
            b[k + 1] -= m * b[k]
        end
        if replace_row > k
            m = A[replace_row, k] / A[k, k]
            A[replace_row, k] = 0.0
            for j in k + 1:n
                A[replace_row, j] -= m * A[k, j]
            end
            b[replace_row] -= m * b[k]
        end
    end
    pi = zeros(n)
    pi[n] = b[n] / A[n, n]
    for i in n-1:-1:1
        sum_ax = 0.0
        for j in i + 1:n
            sum_ax += A[i, j] * pi[j]
        end
        pi[i] = (b[i] - sum_ax) / A[i, i]
    end
    pi
end
```

2. For better or for worse

Consider $A \in \mathbb{R}^{m \times n}$ where $m > n + 1$. Argue that for any $b \in \mathbb{R}^m$, the singular values of A interlace the singular values of $[A \quad b]$. Argue that therefore $\kappa_2([A \quad b]) \geq \kappa_2(A)$.

Let $A \in \mathbb{R}^{m \times n}$ with $m > n + 1$, and let $b \in \mathbb{R}^m$ be arbitrary. Denote $B = [A \quad b] \in \mathbb{R}^{m \times (n+1)}$. The singular values of A are $\sigma_1^A \geq \sigma_2^A \geq \cdots \geq \sigma_n^A \geq 0$, and those of B are $\sigma_1^B \geq \sigma_2^B \geq \cdots \geq \sigma_{n+1}^B \geq 0$.

The Gram matrix $A^\top A$ is symmetric positive semidefinite of size $n \times n$, with eigenvalues $(\sigma_k^A)^2$. Similarly, $B^\top B = \begin{bmatrix} A^\top A & A^\top b \\ b^\top A & b^\top b \end{bmatrix}$ is symmetric positive semidefinite of size $(n+1) \times (n+1)$, with eigenvalues $(\sigma_k^B)^2$. Here, $A^\top A$ is the leading $n \times n$ principal submatrix of $B^\top B$. By the Cauchy interlacing theorem for eigenvalues of Hermitian matrices, if $\lambda_1 \geq \cdots \geq \lambda_{n+1}$ are the eigenvalues of $B^\top B$ and $\mu_1 \geq \cdots \geq \mu_n$ are those of $A^\top A$, then

$$\lambda_k \geq \mu_k \geq \lambda_{k+1}, \quad k = 1, \ldots, n$$

Since the square root function is monotonically increasing and nonnegative, the same inequalities hold for the singular values:

$$\sigma_k^B \geq \sigma_k^A \geq \sigma_{k+1}^B, \quad k = 1, \ldots, n.$$

This establishes the interlacing property.

The 2-condition number is $\kappa_2(A) = \sigma_1^A / \sigma_n^A$ (assuming full column rank; otherwise infinite). Similarly, $\kappa_2(B) = \sigma_1^B / \sigma_{n+1}^B$. From interlacing, $\sigma_1^B \geq \sigma_1^A$ and $\sigma_{n+1}^B \leq \sigma_n^A$, and then thus, $\kappa_2(B) = \frac{\sigma_1^B}{\sigma_{n+1}^B} \geq \frac{\sigma_1^A}{\sigma_n^A} = \kappa_2(A)$

3. Consider constraint

Consider the problem of minimizing $\rho_A(x)$ subject to the constraint that x is elementwise non-negative ($x_i \geq 0$ for all i). Show that if x^* is the solution to this problem, then $\rho_A(x^*) \leq \lambda_{n-k+1}$ where $\lambda_1 \leq \lambda_2 \leq \ldots \leq \lambda_n$ are the eigenvalues of A and $k = \|x^*\|_0$ denotes the number of nonzero entries of x^*.

Let x^* be the minimizer of $\rho_A(x)$ over $x \geq 0$, $x \neq 0$, with support set S of size $k = \|x^*\|_0$. Restrict to the coordinate subspace corresponding to S: let $y^* > 0$ be the normalized positive vector on S such that $\rho_A(x^*) = (y^*)^T A_S y^*$, where A_S is the $k \times k$ Hermitian principal submatrix of A on indices S.

Since x^* minimizes over the nonnegative orthant and is interior to the face defined by S, y^* minimizes the Rayleigh quotient over the full subspace (no sign constraints in that relative interior). Thus, $\rho_A(x^*)$ is the smallest eigenvalue μ_1 of A_S, with y^* the corresponding eigenvector.

By the Cauchy interlacing theorem for Hermitian matrices, the eigenvalues $\mu_1 \leq \cdots \leq \mu_k$ of A_S satisfy $\lambda_i \leq \mu_i \leq \lambda_{i+n-k}$ for $i = 1, \ldots, k$. In particular, $\mu_1 \geq \lambda_1$ (min-max principle), but for the upper bound on the smallest, $\mu_1 \leq \lambda_{n-k+1}$. Hence, $\rho_A(x^*) = \mu_1 \leq \lambda_{n-k+1}$.